

## Agentic Authority Architecture: Trust Domains, Controls, and the Authority Ladder

## The governing rule I

Every architecture in this review answers the exploitation path defined in sec. 3: harden components so that a compromise of one does not become a compromise of all. The authorized-misuse path needs a different instrument. Its governing rule is simple to state and expensive to honor: **a component exposed to untrusted input must not simultaneously hold broad authority over valuable assets.** A browser parses hostile web content, an agent reads hostile repositories and documents, a parser processes attacker-shaped files — and each of those components is, in the agent era, a candidate for total persuasion rather than mere compromise. The rule is the classical protection problem of (Lampson 1974) restated for agents: the question is never only whether a component can be broken, but who may exercise which authority through it. It is also the confused-deputy hazard of (Hardy 1988) made continuous: the deputy now reads its instructions from the

## The governing rule I

same channel the attacker writes to.

The rule converts directly into architecture. Split the system so that no component both touches untrusted input and holds broad authority; place the authority that must exist — credentials, approvals, deployment — behind interfaces too narrow for a persuaded component to abuse. This section defines that architecture: 7 trust domains, 9 controls that hold regardless of host distribution, and an authority ladder that separates what an agent may do alone from what requires a principal outside it.

## The governing rule I

The authority ladder orders an agent-initiated change through six rungs; agent exercise without external authorization is forbidden:

*propose*  $\prec$  *stage*  $\prec$  *authorize*  $\prec$  *exercise*  $\prec$  *audit*  $\prec$  *revoke*

(1)

The section proposes a design; it is not a claim about the defaults of any distribution reviewed elsewhere in this document. What sharpens the proposal is that the control vocabulary below is no longer speculative: the major shipping coding agents now implement versions of these controls against the same operating-system primitives this review's server candidates deploy, and the convergences — and divergences — are documented in vendor engineering materials that this section cites as evidence of mechanism, not as security results (Anthropic 2026b; OpenAI 2026).

## Seven trust domains I

# Seven trust domains I

Table 1: The seven trust domains of the proposed agentic authority architecture, with the intended contents of each and the restrictions that must be preserved for the domain's boundary to mean anything.

| Trust domain                  | Intended contents                                                                               | Restrictions to preserve                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Administration</b>         | OS management, policy changes, trusted update operations.                                       | No routine browsing, repository builds, document parsing, or agent-generated commands without independent review.                                      |
| <b>Personal identity</b>      | Sensitive browser sessions and personal accounts.                                               | Not colocated with untrusted development dependencies or a general-purpose autonomous agent.                                                           |
| <b>Credential service</b>     | Non-exportable keys where feasible; narrowly scoped signing and credential-issuance operations. | Expose specific operations rather than raw secrets or an unrestricted shell; require approval from outside the agent for consequential operations.     |
| <b>Agent execution</b>        | One task or repository, temporary working files, tightly scoped tools.                          | Disposable VM or comparable isolation; no host home directory, broad credential directory, privileged container socket, or ambient production session. |
| <b>Browsing and intake</b>    | Untrusted websites, downloads, email attachments.                                               | Kept separate from administration and signing; anything crossing into trusted work is explicitly reviewed.                                             |
| <b>Release and deployment</b> | Reviewed build outputs; narrowly authorized deployment actions.                                 | The agent may propose a change but must not be able to alter the approval policy or approve its own release.                                           |
| <b>Recovery</b>               | Known-good configuration, independent backups, recovery credentials.                            | Kept beyond the destructive authority of the daily workstation and its agent.                                                                          |

## Seven trust domains II

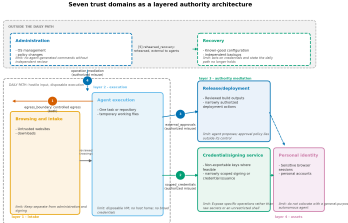


Figure 1: Seven trust domains for AI-assisted work, arranged from hostile intake to protected assets; numbered arrows mark the control catalog entries that mediate each crossing.

## Seven trust domains I

The table and fig. 1 encode two design judgments. First, hostile-input domains and authority domains are different kinds of places: agent execution and browsing/intake are deliberately cheap to destroy, while administration, identity, and credentials are deliberately expensive to enter. Second, the mapping onto concrete platforms varies — for a Qubes deployment these roles become a manageable number of qubes with narrow qrexec policies (Project 2026b, 2026a); for a conventional workstation the highest-risk execution moves to the separate VM host or machine described in sec. 8. The objective in both cases is the same: real separation of authority, not a diagram with many boxes.

## Nine controls and their design rationale I

The domains define where authority lives; the controls define how work crosses the boundaries. Each control below is stated as an operating rule with the design rationale that makes it non-negotiable. Where a shipping agent tool already implements a version of a rule, the rationale cites the documented implementation — as mechanism evidence, not as an endorsement of any tool's security. The nine are also a composed defense rather than a stack of independent toggles: Defense Composition Algebra treats layered controls as formal objects whose composition — coverage interactions included, not just their count — determines what an attacker must defeat (Friedman 2026). In its notation **((def:defense\_composition?))**, a defense over the eight mitigation classes is a composition of per-class layers, written as in

## Nine controls and their design rationale I

eq. 2, and the set below is chosen so that each control closes a failure the others leave open.

Layered controls compose over the eight mitigation classes as a single formal object whose coverage interactions — not merely the count of layers — determine what an attacker must defeat:

$$D = d_1 \circ d_2 \circ \dots \circ d_8 \quad (2)$$

Layered controls interact: a boundary can be bypassed through a dependency of another layer, so adding a control does not monotonically increase what an attacker must defeat. The composition of (**def:defense\_composition?**) bounds the composed whole rather than licensing each layer separately, and it is the audit rung of (**def:authority\_ladder\_order?**) — independent verification against records the agent cannot rewrite — that keeps the composition

## Nine controls and their design rationale I

honest in practice.

Each control above counters a named failure path, and the capability-mediation linkage makes the mapping explicit. Six agent capability classes — content intake, tool and bridge use, credential touch, external communication, state mutation, and self-modification — map onto the failure paths that most directly exploit them and the mediation points that counter them (the mediation taxonomy is developed in sec. 13): content intake and external communication sit on both failure paths, exploitation and authorized misuse, while the other four are exercised chiefly through authorized misuse; sandbox primitives and classifier escalation counter content intake; tool annotations, resource-server binding, and external approval counter tool and bridge use; agent identity exchange and external approval counter credential touch; egress proxy and sandbox observability counter external communication; external approval and

## Nine controls and their design rationale I

tool-bridge constraining counter state mutation; and external approval with independent audit counters self-modification. The linkage is a residual-risk statement as much as a control plan — prompt injection against content intake remains probabilistic under every mediation point — and the composition-interaction caveat stated above is the reason the composition must be read as a whole: the failure paths cross layers, so a dependency of one control can route around the boundary of another.

# Nine controls and their design rationale I

Table 2: The nine controls of the agentic authority architecture: operating rule and design rationale for each. The rules hold regardless of host distribution; the rationales connect each control to a failure it prevents, and where shipping agent tools implement a version of the rule, the rationale points at the documented mechanism.

| Control                    | Operating rule                                                                                                                                                           | Design rationale                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task-scoped credentials    | Prefer short-lived, repository- or service-specific credentials; never hand the agent the owner’s general cloud identity because a task occasionally needs a deployment. | An agent can only misuse authority it holds. Scoping bounds the authorized-misuse blast radius at issuance time, which is the last moment the choice is fully under the principal’s control.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Boundary-controlled egress | Allow only required destinations, at a boundary the agent cannot rewrite, and account for permitted destinations that themselves accept uploads or messages.             | A hostname allowlist is not a data-loss policy: issue trackers, paste sites, CI logs, and webhooks are all “allowed hosts” that exfiltrate. Egress control must govern channels, not merely names. Shipping implementations confirm the mechanism is standard — Claude Code routes sandbox network access through a proxy that enforces an allowed-domains list, and documents the caveats: the proxy mediates by host, dynamic DNS and request-URL discrepancies need explicit attention, and allowed sites that accept arbitrary uploads remain channels (Anthropic 2026b); Codex disables network access by default in its workspace-write mode and requires it to be enabled per policy (OpenAI 2026). Neither mechanism converts a hostname allowlist into a data-loss policy, which is why the control governs channels first. |

# Nine controls and their design rationale II

## External approvals

Require an independent human or deterministic policy check for production changes, secret access, account administration, and publishing.

A second AI instance reading the same hostile material is not automatically an independent security boundary: shared context material and correlated failure modes mean the “reviewer” can be persuaded by the same bytes. Independence must come from different information and authority, not from a second instance. Shipping approval mechanics confirm the rule’s shape: agent tools expose explicit permission modes — default-prompt, accept-edits, plan, and dangerous bypass variants — where the default remains asking the user before consequential action, and Claude Code’s auto mode adds a classifier that judges whether a proposed action is safe enough to proceed without asking (Anthropic 2026b, 2026a). A classifier at the approval rung is an automation of triage, not a replacement for the independent check the rung names. Vague intentions are unenforceable because nothing checks compliance against them. Mediated operations give the approver a concrete object to accept or reject and give the audit trail something to record; this is the confused-deputy fix of (Hardy 1988) applied at the API surface.

## Operation mediation, not vague intentions

Services expose constrained operations — “sign this exact artifact for this release” — never “use the signing key responsibly.” Approval context includes the artifact, destination, scope, and expiration.

Shared state is the channel by which hostile content reaches authority. Every unreviewed promotion is a boundary crossing performed by the

## Minimal shared state

Transfer only the files a task needs; treat returned code, documents, and build artifacts as untrusted until checked; never automatically promote

# Nine controls and their design rationale III

|                                 |                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Environment refresh</b>      | Destroy task environments when work ends; apply updates before creating the next ones. Distinguish deletion of an environment from revocation of credentials it could access.                               | Refresh caps how long a foothold persists, but deletion touches only the environment: credentials already issued outlive it and must be revoked on their own schedule. Conflating the two leaves live authority attributed to a dead machine.                      |
| <b>Tool-bridge constraining</b> | Review filesystem, browser, repository, CI, cloud, and messaging integrations as explicit grants of authority, scoped like any other credential.                                                            | An isolated process with a powerful API token is still a powerful actor: the tool bridge is where isolation boundaries and API authority meet, and unreviewed bridges are the agent-era equivalent of an unconfined daemon holding a root key.                     |
| <b>Independent audit trail</b>  | Record tool use, permission grants, policy changes, and deployment decisions outside the execution environment; make the record sufficient to reconstruct what happened.                                    | The record must survive the thing being audited and must capture actions, not agent prose. An audit trail inside the environment the agent controls is a draft the agent can edit.                                                                                 |
| <b>Rehearsed recovery</b>       | Test rebuilding the environment, revoking credentials, recovering keys, and restoring data — before the incident. Assume a rebuild cannot undo data already exfiltrated or remote changes already accepted. | Recovery rehearsal converts a theoretical plan into a measured capability. Its honest limit is temporal: restoration acts on the future; exfiltration and accepted remote changes are in the past and are addressed only by revocation and downstream containment. |

---

## Nine controls and their design rationale I

Two further trust-boundary decisions concern the models themselves. For **local inference**, compute is not authority: do not attach sensitive credentials to an inference environment merely because it needs substantial GPU resources. A reasonable design separates the model service from agent execution and from credential mediation, with explicit authentication between them — the model service is a component, not a trusted co-resident. For **hosted models**, the decision to transmit private code, documents, or credentials to a provider is an independent trust-boundary decision that no operating-system choice resolves: it determines who else processes the material, under what retention, under which jurisdiction. Both decisions sit above the host-distribution comparisons of sec. 5, sec. 6, and sec. 7; neither can be settled by switching distributions.

## Convergent implementations in shipping agent tools I

The controls above are sometimes read as a bespoke architecture this review invented. The engineering record says otherwise: three widely deployed coding agents converge on the same primitive set, assembled differently per platform, from a single policy model.

Claude Code documents a filesystem sandbox built on bubblewrap on Linux and Seatbelt on macOS, network access mediated by a proxy that enforces an allowed-domains list, and approval machinery in which every command defaults to asking the user, with explicit modes for accepting edits and a classifier-driven auto mode for lower-risk flows; its documentation also names the allowlist's limits — host-level mediation, DNS dynamics, sites that accept uploads — rather than overstating what a domain list buys (Anthropic 2026b, 2026a). Codex documents the same

## Convergent implementations in shipping agent tools I

convergence from the policy-model side: one SandboxPolicy compiles to Seatbelt on macOS, to bubblewrap and Landlock on Linux, and to Windows DACL/appcontainer mechanisms, with mode presets (read-only, workspace-write, danger-full-access) and network off by default in the write modes (OpenAI 2026). Gemini CLI ships configurable sandboxing across Seatbelt, Docker, gVisor, and bubblewrap backends on its supported platforms (Google 2026). The pattern is the point: three independent vendors, one shared conclusion that agent containment is assembled from the same kernel-level primitives — namespace and MAC-framework sandboxing on the filesystem axis, a mediation point on the network axis, and an approval or classifier layer on the decision axis — that the server candidates of sec. 8 expose as ordinary configuration. The convergence also marks the frontier of the architecture: none of these mechanisms, as documented, includes the

## Convergent implementations in shipping agent tools I

credential-scoping or audit controls of this section, which is exactly the gap the trust-domain design addresses.

The convergence has a design consequence the rationales above now state explicitly. Because the primitives are shared, an operator can evaluate any agent-sandbox claim directly: which filesystem mechanism (bubblewrap, Seatbelt, Landlock), whether egress is proxy-mediated or merely filtered, whether the network default is closed, and which approvals remain outside the agent. Vague answers to those four questions identify an agent whose containment is a convenience, not a boundary.

## The authority ladder I



# The authority ladder I

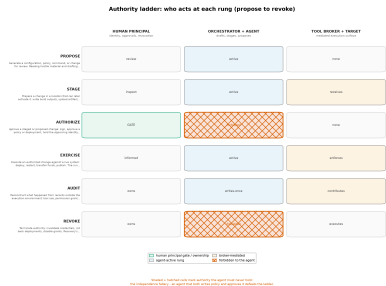


Figure 2: The six-rung authority ladder as a swimlane across human principal, orchestrator and agent, and tool broker; hatched cells mark exercise paths an agent must never hold without external authorization.

## The authority ladder I

The ladder in fig. 2 gives the architecture its operating rhythm across six rungs: **propose**, **stage**, **authorize**, **exercise**, **audit**, **revoke**, ordered as in eq. 1. An agent may propose and stage freely — generating configurations, preparing artifacts, drafting deployments — because nothing at those rungs changes the world. Authorization is the hinge: it must be exercised outside the agent, by the human principal or by deterministic policy, and the approval context is a mediated operation with artifact, destination, scope, and expiration. Exercise happens under whatever scopes the authorization granted, no more. Audit and revoke are not afterthoughts but standing rungs: the audit record lives outside the execution environment so it survives the environment's destruction, and revocation

## The authority ladder I

is the only rung that addresses what a rebuild cannot — authority already exercised against the outside world. An architecture that lets the agent climb from propose to authorize on its own has not built a ladder; it has built a loop, and the configuration-side invariants that prevent exactly that are specified and enforced in sec. 14. The permission modes of the shipping tools read as partial ladders: default-prompt modes place authorize with the human; accept-edits moves routine edit-authorization to the agent while keeping command approval; bypass modes remove the hinge altogether and are, in this architecture's terms, a design admission rather than a mode (Anthropic 2026b; OpenAI 2026).

## The authority ladder I

The authorize/exercise split of (**def:authority\_ladder\_order?**) is also what keeps delegated trust bounded, and it has a commercial-integrity ancestry: Clark and Wilson grounded integrity in well-formed transactions, in which data changes state only through a mediated operation whose form is certified in advance, and in separation of duty, which prevents the principal who certifies an operation from being the one who executes it (Clark and Wilson 1987). Operation mediation and the approve/exercise split of this section are those controls restated for agents. Formal treatments of agent trust bound a delegate's effective trust by what it was authorized to exercise rather than by what it can reach, and the ladder encodes that bound structurally: authorization fixes the scope, exercise consumes it, and no rung is reachable that lets the agent enlarge its own authorization (Friedman 2026). The orchestration section extends the same bound across

## The authority ladder I

agent-to-agent delegation chains, where unbounded trust would otherwise amplify with each hop sec. 13.

The trust domains, controls, and ladder compose into the scenarios of sec. 16: a Qubes deployment realizes the domains as qubes with narrow communication policies (Project 2026b), a conventional workstation realizes them through the separate execution infrastructure of sec. 8, and the orchestration layer realizes the ladder over fleets as described in sec. 13. The operator behaviors that keep the separation real — and the cognitive failure modes that erode it — are the subjects of sec. 12 and sec. 11.